Diagnosis and Identification of Exceptional Learners

Learners with disabilities, the gifted and the twice-exceptional



Dr. Omar A. Alsamani

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Website: alsamani.com

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Appendix A

Guide to instruments

A concise reference guide to the instruments presented in this book (54 tools and procedures), grouped by purpose, with a pointer to where each is discussed in the chapters.

Before using this guide

This appendix brings together the instruments presented in the book's instrument profiles and in its tables of instruments and their Arabic versions, arranged in eight tables by purpose. For each instrument it records only what the book itself reports: purpose, age range, examiner qualification level and the status of its Arabic version, followed by the chapter in which it is presented. Age ranges, qualification levels and Arabic versions are all given as they stood when the book was prepared; they change as new editions, new Arabic standardizations and new publisher requirements appear, so the publisher and the instrument's own manual should be consulted before any professional use. This guide is no substitute for the full instrument profile in the relevant chapter.

Purpose:

- Screening
- Rating
- Diagnosis
- Identification

Monitoring

. A dash (—) means the book does not give this information for the instrument. An asterisk (*) marks an instrument mentioned in the body of a chapter rather than in a profile or table. The chapter number points to the profile or table in which the instrument first appears in that chapter.

A.1 Intelligence and abilities

Table A.1: Individual intelligence and ability tests and nonverbal ability tests.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Wechsler Intelligence Scale for Children, Fifth Edition WISC-V	Diagnosis	6 to 16 years 11 months	—	We know of no published Saudi standardizatio n of the fourth or fifth edition. The older revised edition (**WISC-R**) was standardized in Saudi Arabia for ages 6–16 (Al-Nafi et al., 2000 [Arabic], as cited in Ejbara, 2024)	**Ch. 3** **Ch. 7** **Ch. 12** **Ch. 13**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Wechsler Preschool and Primary Scale of Intelligence, Third Edition WPPSI-III	Diagnosis	The book does not give the range of the original; the Saudi version was administered from 4 years to 7 years 3 months	—	A Saudi version derived from a Jordanian version, administered to 788 children in the Jeddah education region between 2018 and 2022 (Ejbara, 2024); its generalizability to other regions of the Kingdom has not been established	**Ch. 3** **Ch. 7**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Stanford-Binet Intelligence Scales, Fifth Edition SB5	Diagnosis	2 to 85+ years	Publishers restrict it to those with advanced training in administering and interpreting individual intelligence tests; check the publisher's current designation for the level	Two Gulf versions are in use in the Kingdom, one prepared by Mahmoud Abu Al-Nil and colleagues and the other by Safwat Farag; a study in the Northern Borders region compared them (Mohammed, 2016 [Arabic]). We found no published data on their norming samples or the year the data were collected	**Ch. 3** **Ch. 7** **Ch. 11**
Kaufman Assessment Battery for Children, Second Edition, Normative Update KABC-II NU	Diagnosis	3 to 18 years	—	No published standardized Arabic version is known	**Ch. 3**
Differential Ability Scales, Second Edition DAS-II	Diagnosis	2 years 6 months to 17 years 11 months	—	No published standardized Arabic version is known	**Ch. 3**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Cognitive Assessment System, Second Edition CAS2	Diagnosis	5 to 18 years 11 months	—	No published standardized Arabic version is known	**Ch. 3**
Leiter International Performance Scale, Third Edition Leiter-3	Diagnosis	3 to 75+ years	—	No published standardized Arabic version is known	**Ch. 3** **Ch. 11**
Universal Nonverbal Intelligence Test, Second Edition UNIT2	Diagnosis	5 to 21 years 11 months	—	No published standardized Arabic version is known	**Ch. 3** **Ch. 11** **Ch. 13**
Test of Nonverbal Intelligence, Fourth Edition TONI-4	Screening	6 to 89	—	No published standardized Arabic version is known	**Ch. 3**
Raven's Progressive Matrices Raven's	Screening	A wide range, depending on the form	—	No published Arabic standardizatio n of its current forms has been verified	**Ch. 3**

A.2 Achievement and academic skills

Table A.2: Comprehensive achievement batteries and tests of specific skills in reading, mathematics and writing.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Wechsler Individual Achievement Test, Fourth Edition WIAT-4	Diagnosis	4 to 50 years 11 months	An examiner trained in administering and interpreting individually administered standardized tests, in accordance with the publisher's requirements	No published standardized Arabic version is known	**Ch. 3** **Ch. 8**
Kaufman Test of Educational Achievement, Third Edition KTEA-3	Diagnosis	4 to 25 years 11 months	—	No published standardized Arabic version is known	**Ch. 3**
Woodcock-Johnson IV Tests of Achievement WJ IV ACH	Diagnosis	2 to 90+ years	—	No published standardized Arabic version is known. The publisher refers to a fifth edition (**WJ V**), so check for the most recent edition	**Ch. 3**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
KeyMath-3 Diagnostic Assessment KeyMath-3	Diagnosis	4 years 6 months to about 21 years	An examiner trained in administering and interpreting individually administered standardized tests, in accordance with the publisher's requirements	No published standardized Arabic version is known	**Ch. 8**
Test of Early Mathematics Ability, Third Edition TEMA-3	Diagnosis Screening	3 years to 8 years 11 months	An examiner trained in administering and interpreting individually administered standardized tests, in accordance with the publisher's requirements	No published standardized Arabic version is known, nor any standardized Arabic instrument for number sense	**Ch. 8**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Test of Written Language, Fourth Edition TOWL-4	Diagnosis	9 years to 17 years 11 months	An examiner trained in administering standardized tests and scoring writing samples, in accordance with the publisher's requirements	No published standardized Arabic version is known	**Ch. 8**
Comprehensive Test of Phonological Processing, Second Edition CTOPP-2	Diagnosis	4 to 24 years, in two age-based forms	An examiner trained in administering and interpreting individually administered standardized tests, in accordance with the publisher's requirements	No published standardized Arabic version is known. Research has used Arabic tasks built on the same idea (Taibah & Haynes, 2011), but they may not be interpreted against the original norms	**Ch. 8** **Ch. 13**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Test of Word Reading Efficiency, Second Edition TOWRE-2	Diagnosis Monitoring	6 to 24 years	An examiner trained in administering and interpreting individually administered standardized tests, in accordance with the publisher's requirements	No published standardized Arabic version is known. The closest we found is a Jordanian measure of word recognition fluency for monitoring first graders (Abu-Hamour, 2014b), which is a monitoring tool, not a diagnostic test	**Ch. 8**
Curriculum-based measures: oral reading fluency, maze, math computation and written expression CBM	Screening Monitoring	Passages and problems drawn from the student's grade-level curriculum	—	Arabic studies of curriculum-based reading fluency measures (Jordan; United Arab Emirates) with small samples; they do not provide norms (Abu-Hamour, 2014a; Mahfouz & Mohamed, 2023)	**Ch. 4** **Ch. 8**

A.3 Adaptive behavior

Table A.3: Adaptive behavior scales used in diagnosing intellectual disability and determining its severity.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Vineland Adaptive Behavior Scales, Third Edition Vineland-3	Rating Diagnosis	Birth to 90 years for the interview and parent form; 3 to 21 years for the teacher form	The interview requires an examiner trained in semi-structured interviewing; check the level specified by the publisher	Arabic catalogs list a title, “Vineland Adaptive Behavior Scale: Saudi Norms,” but its author, year and sample, and the edition on which it was based, could not be verified. No published Arabic standardizatio n of the third edition is known	**Ch. 3** **Ch. 7**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Adaptive Behavior Assessment System, Third Edition ABAS-3	Rating Diagnosis	Birth to 89 years, depending on the form	Completed by a parent or teacher; scored and interpreted by a qualified professional; check the level specified by the publisher	The teacher form was adapted into Arabic in Oman and studied with 410 children with intellectual disability; the study supported a single general factor and high reliability (Emam et al., 2020). No published Saudi standardizatio n is known	**Ch. 3** **Ch. 7**
Diagnostic Adaptive Behavior Scale DABS	Diagnosis	4 to 21 years	—	Nothing published was found	**Ch. 7**

A.4 Development and early screening

Table A.4: Developmental screening and developmental assessment instruments in early childhood.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Ages & Stages Questionnaires, Third Edition ASQ-3	Screening	Questionnaire intervals from 2 to 60 months; usable from 1 to 66 months	Completed by parents; scored and interpreted by a professional trained in early childhood or health care; no psychological assessment license required	No published standardized Arabic version is known, and translation alone does not carry over the U.S. cutoff scores	**Ch. 7**
Bayley Scales of Infant and Toddler Development, Fourth Edition* Bayley-4	Diagnosis	16 days to 42 months	—	No published standardized Arabic version is known	**Ch. 3** **Ch. 5**

A.5 Autism

Table A.5: Screening, rating and diagnostic instruments for autism spectrum disorder.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Modified Checklist for Autism in Toddlers, Revised with Follow-Up M-CHAT-R/F	Screening	16 to 30 months	Completed by parents; the follow-up interview is conducted by a physician, nurse or specialist trained in it; no psychological assessment license required	The first version was studied in nine Arab countries with 228 children: sensitivity 0.86, specificity 0.80 (Seif Eldin et al., 2008). Approved translations of the revised version are available on the official website, but no published Arabic standardizatio n study of it is known	**Ch. 9**
Social Communication Questionnaire SCQ	Screening	4 years and older, provided mental age exceeds 2 years	Completed by a parent; scored and interpreted by a trained professional in accordance with the publisher's requirements	No published standardized Arabic version is known	**Ch. 9**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Social Responsiveness Scale, Second Edition SRS-2	Rating	2 years 6 months to adulthood	—	No published standardized Arabic version is known	**Ch. 9** **Ch. 13**
Childhood Autism Rating Scale, Second Edition CARS-2	Rating Diagnosis	2 years and older	A specialist trained in autism assessment, after observing the child and gathering information about the child	An Arabic edition prepared by Leila Akoury-Dirani and Mayssam Alameddine (Akoury-Dirani & Alameddine, 2016 [Arabic]), for which no published description of the sample was found; a factor-structure study with 301 children (Alotaibi & Alotaibi, 2021); and an adapted Iraqi version (Al-Shibly et al., 2025). No published Arabic norms for the T-scores are known	**Ch. 9**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Gilliam Autism Rating Scale, Third Edition GARS-3	Rating	3 to 22 years	Completed by parents, a teacher or a specialist; interpreted by a qualified specialist	An Arabic version studied in Saudi Arabia with 178 participants, with a Cronbach's alpha of 0.971 (Alasmari et al., 2024a, 2024b). No published Saudi norms for the index are known, so scores are read against the U.S. norms	**Ch. 9**
Autism Diagnostic Interview–Revised ADI-R	Diagnosis	Individuals with a mental age above 2 years	A specialist with formal training in administering and coding the interview	No approved or published standardized Arabic version is known	**Ch. 9**
Autism Diagnostic Observation Schedule, Second Edition ADOS-2	Diagnosis	12 months to adulthood, in five modules	A specialist with formal training in the instrument; the publisher restricts purchase to those with advanced qualifications	No Arabic translation approved by the publisher is known, nor any published standardized Arabic version	**Ch. 9**

A.6 Attention, behavior and emotion

Table A.6: Broadband screening tools, narrowband and broadband rating scales, and universal behavioral screening systems in schools.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Strengths and Difficulties Questionnaire SDQ	Screening	Parent and teacher forms from 2 to 17 years, with age-specific versions; a self-report form from 11 to 17 years	Completed by parents, teachers and students; interpreted by a trained professional	An Arabic translation is available on the official website; its validity was examined with Yemeni children aged 5 to 12 in a clinical sample (87) and a community sample (100) (Alyahri & Goodman, 2006). No published Saudi norms are known	**Ch. 3** **Ch. 10**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Vanderbilt ADHD Diagnostic Rating Scales VADRS	Rating	Primarily elementary school (about 6 to 12 years); the Saudi study included children aged 5 to 12	Completed by parents and teachers; interpreted by a physician or qualified specialist	An Arabic version (**VADRS-A**) with parent and teacher forms, studied in Saudi Arabia with 1,812 raters (504 parents and 1,308 teachers) of children aged 5 to 12 (Alqahtani et al., 2024)	**Ch. 3** **Ch. 10**
Conners 4th Edition Conners 4	Rating	6 to 18 years for the parent and teacher forms, with a self-report form for older youth	A professional trained in interpreting standardized rating scales, in accordance with the publisher's requirements	No standardized Arabic version published in a peer-reviewed journal is known	**Ch. 3** **Ch. 10**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Achenbach System of Empirically Based Assessment: Child Behavior Checklist (CBCL), Teacher's Report Form (TRF) and Youth Self-Report (YSR) ASEBA	Rating	6 to 18 years for the parent and teacher forms and 11 to 18 years for the self-report; separate forms for preschoolers	Completed by parents, teachers and students; interpreted by a professional trained in standardized rating scales	No published standardized Arabic version with local norms is known; ask the publisher about licensed translations	**Ch. 3** **Ch. 10**
Behavior Assessment System for Children, Third Edition BASC-3	Rating	2 to 21 years 11 months for the parent and teacher forms	—	No published standardized Arabic version is known	**Ch. 3** **Ch. 10**
Revised Child Anxiety and Depression Scale RCADS	Rating Screening	Students in roughly Grades 3 through 12	Completed by the student and parents; interpreted by a trained professional	No published standardized Arabic version is known	**Ch. 10**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Computerized continuous performance tests CPT	Diagnosis	Vary by test; most start at school age	A trained examiner administers it and a qualified professional interprets it, in accordance with the publisher's requirements	The task is usually nonverbal, but no published standardized Arabic norms are known for any of these tests; they are a supplementary measure on which a diagnosis cannot rest alone	**Ch. 10**
Student Risk Screening Scale and its extended version for internalizing problems SRSS · SRSS-IE	Screening	—	The teacher rates every student on a single list; no qualification level is specified	No published standardized Arabic version of universal behavioral screening systems is known	**Ch. 10**
Behavioral and Emotional Screening System BESS	Screening	—	Brief teacher, parent and student forms; no qualification level is specified	No published standardized Arabic version of universal behavioral screening systems is known	**Ch. 10**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Systematic Screening for Behavior Disorders SSBD	Screening	—	The teacher ranks the class and rates the highest-ranked students, followed by direct observation in the original version; no qualification level is specified	No published standardized Arabic version of universal behavioral screening systems is known	**Ch. 10**

A.7 Hearing, vision and communication

Table A.7: Procedures for assessing hearing and functional vision, and language tests.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Pure-tone audiometry, tympanometry and objective measures (otoacoustic emissions and auditory brainstem response) PTA · OAE · ABR	Screening Diagnosis	Objective measures from birth; behavioral testing with age-appropriate methods from about 6 months, and by the conventional method from about age 5	A licensed audiologist; school screening is conducted by trained personnel under professional supervision	Pure-tone audiometry requires no language. Speech discrimination testing, however, requires equivalent Arabic word lists, and no recent published standardized Saudi lists have been verified	**Ch. 11**
Functional vision assessment and learning media assessment FVA · LMA	Rating	All ages; LEA Symbols and four-letter charts (**HOTV**) are used with young children from about age three	A teacher of students with visual impairments, together with an orientation and mobility specialist	LEA Symbols do not depend on language. No published Arabic guide to learning media assessment is known, so graded Arabic texts are prepared locally	**Ch. 11**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Clinical Evaluation of Language Fundamentals, Fifth Edition CELF-5	Diagnosis	5 to 21 years 11 months	A speech-language pathologist or qualified examiner, in accordance with the publisher's requirements	No published standardized Arabic version is known, and translating its items does not yield a valid instrument, because the structures it measures are English	**Ch. 11**

A.8 Giftedness and creativity

Table A.8: Group ability tests, rating scales and observation tools, creativity tests, and identification procedures.

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Cognitive Abilities Test, Form 7 CogAT	Screening Identification	Kindergarten through Grade 12, in grade-based levels	Administered in class by the teacher under standardized directions; results interpreted by a trained professional	No published standardized Arabic version is known	**Ch. 3** **Ch. 12**
Naglieri Nonverbal Ability Test, Third Edition NNAT3	Screening Identification	Kindergarten through Grade 12	—	No published standardized Arabic version is known	**Ch. 3** **Ch. 12**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Mawhiba Multiple Cognitive Abilities Test MMCAT	Identification	Grade 3 of elementary school through Grade 10 (first year of secondary school), at three levels	No qualification level is specified; administered by the National Center for Assessment at the Education and Training Evaluation Commission, in partnership with Mawhiba and the Ministry of Education	Originally developed in Arabic for the Saudi context and used nationally since 2011. No published technical manual was found, nor information on its score scale or norming year	**Ch. 3** **Ch. 12**
Gifted Rating Scales GRS	Rating Identification	School Form for Grades 1–8 (ages 6–13); Preschool Form from 4 years to 6 years 11 months	Completed by a teacher who has known the student long enough; interpreted by a professional in accordance with the publisher's requirements	No published standardized Arabic version is known, nor any Saudi standardizatio n study	**Ch. 12**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Scales for Rating the Behavioral Characteristics of Superior Students SRBCSS	Rating Identification	—	Rating scales completed by the teacher; no qualification level is specified	No published Saudi standardizatio n study was found; the scales have no national norms and are read against local norms built by the program	**Ch. 12**
Teacher's Observation of Potential in Students TOPS	Identification	Designed for young students	The teacher observes students over several weeks during enriched learning activities; no qualification level is specified	No published Saudi standardizatio n study was found	**Ch. 12**
Traits, Aptitudes, and Behaviors TABs	Identification	—	Observed by the teacher; no qualification level is specified	No published Saudi standardizatio n study was found	**Ch. 12**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Torrance Tests of Creative Thinking TTCT	Identification	Kindergarten to adulthood	Administered by a teacher or specialist after training; scored by a trained scorer or by the publisher's scoring service	Figural Form B was standardized in Saudi Arabia in 1416 AH with 1,227 male and female students aged 9 to 16 (Al-Nafi, 1428 AH [Arabic]); the original standardization report could not be consulted, and the norms are nearly thirty years old	**Ch. 12**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Above-level testing Above-level	Identification	Typically Grades 3 through 9 in talent search programs	Administered by whoever is qualified to administer the original test under its conditions; interpreted by a professional familiar with the norms of the higher grade	No published Arabic program with reported norms is known. The idea can be applied with Arabic achievement tests for higher grades, provided this is stated explicitly and the result is read against the norms of the grade for which the test was designed	**Ch. 12**

Instrument	Purpose	Ages	Examiner qualification	Arabic version	Chapte
Two-lens assessment protocol for a student suspected of being twice-exceptional 2e	Diagnosis Identification	School age; instruments are selected according to the student's age	A team that includes expertise in both giftedness and disability; individual measures are administered by a qualified psychologist in accordance with each publisher's requirements	No standardized Arabic protocol for twice-exceptional learners is known	**Ch. 13**

When is it sound to conclude that a student has a disability, a disorder, giftedness or twice-exceptionality, and on what grounds?

This book traces the stages of diagnosis and identification systematically, from referral and the gathering of evidence to the interpretation of that evidence and the decisions built on it. It goes beyond defining the categories and setting out their criteria to examine the nature of measurement instruments, the procedures for administering them, the qualifications required of those who administer them, the principles for interpreting their results, and the point at which the evidence is sufficient to support a judgment.

The book brings together two fields that the literature usually treats separately: disabilities and disorders, and giftedness and twice-exceptionality. It reads international standards and the American experience in light of published Saudi regulations, and describes the status of the Arabic version of each instrument it presents.

Its fifteen chapters address the foundations of measurement; formal, informal and alternative methods of gathering evidence; and identification models; then take up the categories of exceptionality one by one, supported by decision pathways, instrument profiles and applied cases. They close with fairness in diagnosis, report writing, and the development of the individualized education program.

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